ZUGFERD-ENTWICKLERTAGE 2026

Syntax Binding of **EN 16931-1:2026**

Semantics is the cargo, syntax the carrier — what the new standard means for UBL, CII and ZUGFeRD.

Slides as PDF

Svante Schubert
Editor of the syntax bindings · CEN/TC 434 WG3 · for DIN

SEMANTICS & SYNTAX

Semantics is the cargo, **syntax** the carrier. In 2026 **the cargo grows** — and **both carriers** are modernised.

Semantics
164 → 216 Business Terms

UBL 2.1 → 2.5
OASIS Standard since 08/2026

CII D16B → D25A
UN/CEFACT, 06/2026

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NEW SEMANTICS · OVERVIEW

What the standard can say now

Three pillars of the revision: Correct errors | B2B obligations of the Member States | Requirements from ViDA

Payment terms

Multiple payment terms; early payment discount and late payment penalty 0..n each — structured instead of free text.

Tax & ViDA

VAT category N, exemption reasons for allowances and charges, subtotals per exemption reason, goods/services.

References per line

Purchase order, despatch advice, receiving advice and delivery note at document and line level.

Delivery per line

Deliver-to party, location, date and address per invoice line.

Third-party fees

Show amounts on behalf of third parties that are not part of the invoice separately.

XML attachments

Embed structured documents as attachments to the invoice.

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SEMANTICS COMPARED

2017 → 2026

The data model grows — above all in the business terms.

BG · Business Groups
Structural groups

2017	32
2026	39

+7 +22%

BT · Business Terms
Numbered main terms

2017	164
2026	216

+52 +32%

Auxiliary attributes
e.g. scheme identifier

2017	19
2026	22

+3 +16%

BOTTOM LINE

+52

new business terms — the largest growth

- 7 new business groups structure the new fields
- Auxiliary attributes grow only moderately (+3)
- **+62 elements in total in the data model**

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SEMANTIC DATA TYPES

Distribution by data type

Total 164 → 216 +52 · +32%

DATA TYPE	2017		2026	CHANGE	IN %
Code	21	→	**37**	+16	+76%
Document reference	10	→	**22**	+12	+120%
Text	62	→	**72**	+10	+16%
Amount	21	→	**25**	+4	+19%
Date	9	→	**13**	+4	+44%
Identifier	27	→	**30**	+3	+11%
Percentage / Percent	8	→	**10**	+2	+25%
Time	0	→	**1**	+1	new
Quantity	2	→	**2**	±0	±0%
Unit price amount	3	→	**3**	±0	±0%
Binary object	1	→	**1**	±0	±0%

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SYNTAX · BACKGROUND

One semantics, three syntaxes

EDIFACT and CII come from the UN family, UBL from OASIS — the Core Components connect the two XML syntaxes.

	UN/EDIFACT	OASIS UBL	UN/CEFACT CII
1987 – 1988	Syntax rules adopted by UN/ECE WP.4; ISO 9735 (1988)	–	–
1990s	INVOIC among the first UN standard messages; UN/CEFACT succeeds WP.4 (1996/97)	–	–
1998 – 2002	Syntax version 4 (ISO 9735:1998): UTF-8 possible via UNOW	UBL TC founded at OASIS (2001), based on ebXML	ebXML Core Components (UN/CEFACT + OASIS, 1999 – 2001)
2004 – 2009	–	UBL 1.0 (2004), UBL 2.0 (2006)	CII v1 based on the UN/CEFACT Core Component Library
2013 – 2016	INVOIC D16B	UBL 2.1 (2013) → ISO/IEC 19845:2015	CII D16B (2016)
2017	CEN/TS 16931-3-4 (**INVOIC D.16B**)	CEN/TS 16931-3-2 (**UBL 2.1**)	CEN/TS 16931-3-3 (**CII D.16B**)
2026	CEN/TS 16931-3-4 (**INVOIC D.24A**)	CEN/TS 16931-3-2 (**UBL 2.5**)	CEN/TS 16931-3-3 (**CII D.25A**)

EN 16931-1:2017 + Implementing Decision (EU) 2017/1870: only UBL 2.1 and CII D.16B are on the list of syntaxes

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SYNTAX · VERSIONS

UBL 2.5 and CII D25A

From UBL 2.1 (2013) and CII D16B (2016) to the 2026 editions.

OASIS UBL

UBL 2.1 → **UBL 2.5** **232 → 304**

Entries in the mapping

- OASIS Standard since 12 August 2026
- Buyer reference gets its own structure and may repeat
- Invoice notes move into their own annotation structure
- 7 paths and 24 cardinalities changed

```
BT-10:  cbc:BuyerReference →
          cac:BuyerAssignedReference/cbc:BuyerReference
```

UN/CEFACT CII

CII D16B → **CII D25A** **241 → 312**

Entries in the mapping

- Published by UN/CEFACT in June 2026
- Buyer reference moves into a new element and may repeat
- VAT category points to the varying code (standard rate, reverse charge, exempt ...)
- 21 paths and 28 cardinalities changed

```
BT-10:  ram:BuyerReference → ram:BuyerReferenceID
BT-95/102/118/151:  ram:TypeCode → ram:CategoryCode
```

In both syntaxes: 26 known errors of the 2017 mapping tables fixed

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SYNTAX · OVERVIEW

The big picture in one sentence

The 2026 invoice can **say more** — and says it **more clearly**.

+30%

more content

CII mapping 241 → 312 entries, UBL 232 → 304

7

new information groups

BG-33 … BG-39

52

new business terms

Data fields 164 → 216

26

errors fixed

known errors of the 2017 tables, per format

Both XML formats on current versions: CII D16B → D25A and UBL 2.1 → 2.5

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FORMATS · EUROPE

E-invoicing formats in the EU

AT ebInterface
National XML 2014 federal B2G

EE EVS 923
National XML 2019 B2G

ES Facturae or UN/EDIFACT
National XML or EDIFACT 2015 B2G

FI Finvoice or TEAPPSXML
National XML 2020 right to e-invoice

IT FatturaPA
National XML 2019 B2B

PL FA(3)
National XML 2026 B2B phased

SI e-SLOG
National XML 2015 B2G

BE Peppol BIS Billing
UBL XML mandatory, CII additionally 2026 B2B

DE XRechnung or ZUGFeRD / Factur-X
UBL/CII XML or PDF + CII XML 2025 B2B receipt

FR Factur-X
PDF + CII XML 2026 B2B phase 1

National XML = own syntax; **green** = built on UBL or CII. Years = legal milestones for the scope named. Sources: links and notes.

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FORMATS · GERMANY & FRANCE

ZUGFeRD & Factur-X

A German-French standard — France has been on board with Factur-X since 2017.

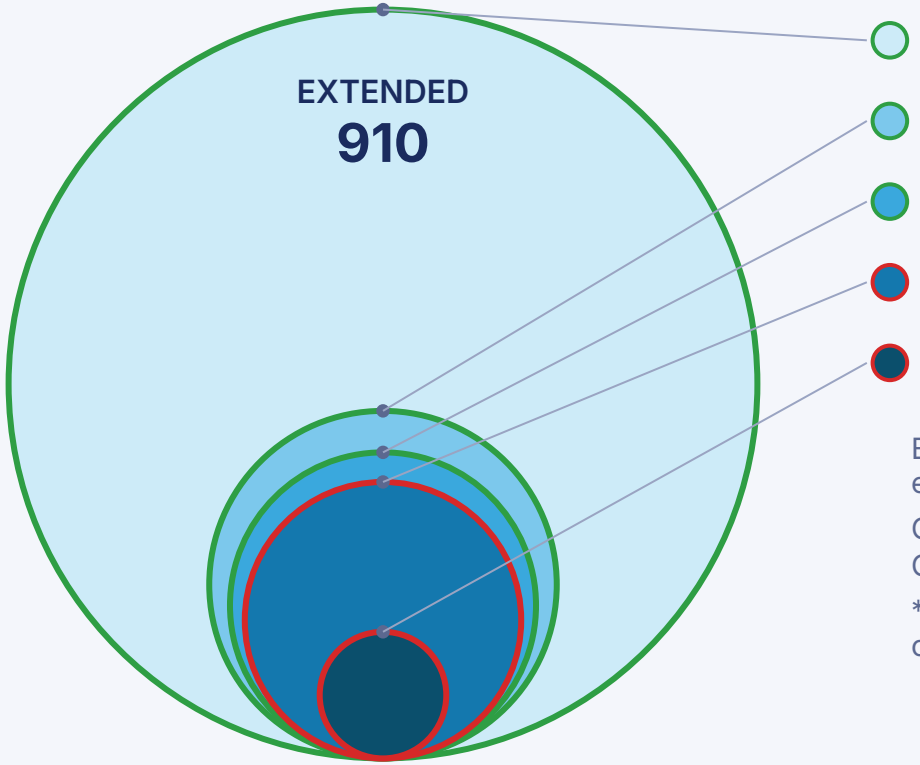
Date	ZUGFeRD	Factur-X	Milestone
2014-06-25	1.0	–	Hybrid invoice of PDF/A-3 + CII — designed before EN 16931
2017-12-31	–	1.0	France launches Factur-X (FNFE-MPE) based on EN 16931
2019-03-11	2.0	1.0.03	ZUGFeRD 2.0 = Factur-X 1.0.03: rebuilt on EN 16931 (CII D16B)
2019-10	2.0.1	1.0.04	Technical alignment — from here on an e-invoice as defined by the BMF
2020-03-24	2.1	1.0.05	Full alignment: one common German-French standard
2020-07	2.1.1	1.0.05	New profile XRECHNUNG — the bridge to the public administration's standard
2022-03-01	2.2	1.0.06	Reference profile XRECHNUNG; EXTENDED for Order-X and the B2B reform in France
2024-09-18	2.3	1.0.07	CII D22B instead of D16B; alignment with XRechnung 3
2025-12-04	2.4	1.08	Sub-lines with subtotals, machine-readable delivery date
2026-06-10	2.5	1.09	IDs of the EN 16931 revision; maintenance 2.5.2 / 1.09.2 on 2026-08-04

Green = e-invoice in Germany (BMF: from ZUGFeRD 2.0.1, without MINIMUM and BASIC WL) · **Red** = not an e-invoice

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FORMATS · ZUGFERD PROFILES

ZUGFeRD 2.5.2 / Factur-X 1.09.2: size of the profiles



Profile	Business Groups (BG)	Business Terms (BT)	Extension (BG-X + BT-X)	Σ	E-invoice DE	E-invoice FR
EXTENDED	33	175	93 + 609	**910**	yes	yes
EN 16931	32	164	–	**196**	yes	yes
BASIC	27	125	–	**152**	yes	no*
BASIC WL	21	103	–	**124**	no	until 09/2027
MINIMUM	8	18	–	**26**	no	no

BG = business group (group of related fields, e.g. BG-4 SELLER); BT = business term (single field, e.g. BT-1 Invoice number).

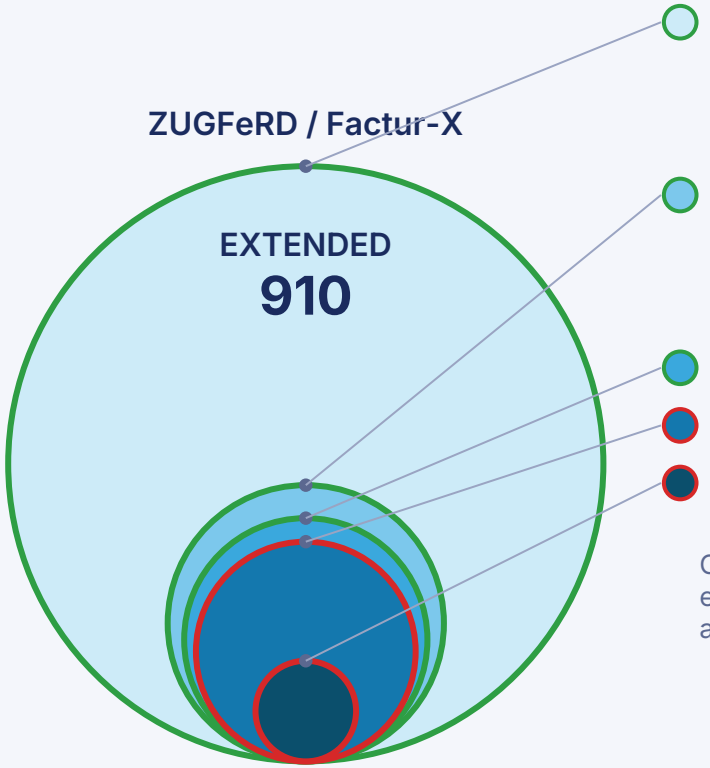
Circle area ∝ number of BG + BT; each profile contains all smaller ones. Green: e-invoice in Germany; red: MINIMUM and BASIC WL are not (BMF).

* FR: BASIC is to be declared as EN 16931 (AFNOR XP Z12-012). EXTENDED already uses 12 new IDs of the revision (BG-34, BT-173 … BT-216).

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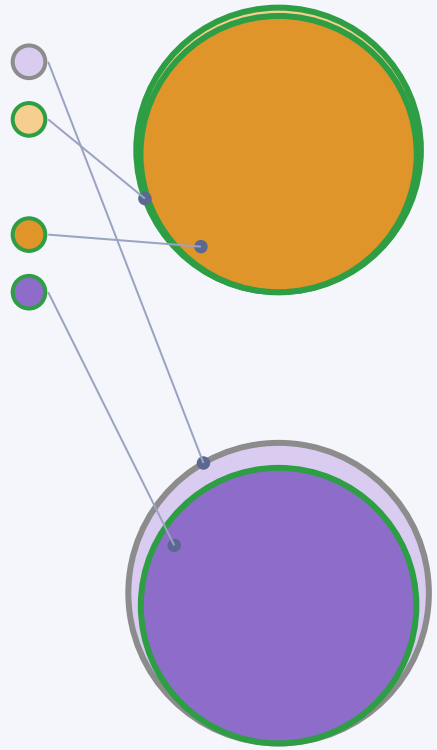
FORMATS · COMPARISON

ZUGFeRD, XRechnung and Peppol compared



Model / profile	Groups (BG)	Terms (BT)	New BG + BT	Σ
EXTENDED	33	175	93 + 609	910
Peppol PINT	31	163	6 + 32	232
XRechnung Ext.	32	164	9 + 3	208
EN 16931	32	164	–	196
XRechnung CIUS	32	164	–	196
Peppol BIS 3.0	32	163	–	195
BASIC	27	125	–	152
BASIC WL	21	103	–	124
MINIMUM	8	18	–	26

Circle area ∝ business groups (BG) + business terms (BT). Outline green = e-invoice in Germany, red = not (MINIMUM, BASIC WL), grey = PINT is not a CIUS of EN 16931 (in Germany only by agreement).



XRechnung 3.0.2
Ext. 208
CIUS 196

Peppol
PINT 232
BIS 3.0 195

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NEW SEMANTICS · CALCULATION RULES

Rounding & tolerance

For the first time the standard specifies how precisely to calculate and compare.

Rounding

All amounts are rounded to the decimal places of the smallest currency unit — the “minor unit” according to ISO 4217.

EXAMPLES decimal places

EUR	Euro	**2**
ISK	Icelandic króna	**0**
JPY	Japanese yen	**0**
KWD	Kuwaiti dinar	**3**

JPY and KWD are outside Europe — no European currency has 3 decimal places.

Tolerance BT-92, 99, 131

Document level allowance amount, Document level charge or tax amount, Invoice line net amount

Decimal places of the currency	Tolerance
0 (e.g. ISK, JPY)	2
1	0.2
2 (e.g. EUR)	0.02
3 (e.g. KWD)	0.002
4	0.0002

MULTIPLICATION & DIVISION

0.1%

of the calculated amount as the largest permitted deviation

- at least 0.01
- at most 1.00
- for all other calculation rules

Tolerances absorb rounding differences — better they never arise: → **Amendment #163**

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EXACT INVOICE ARITHMETIC · 1/5

Computers usually calculate in binary

XSLT, the standard for validating e-invoices, calculates in binary — unless told otherwise.

AN EVERYDAY COMPARISON

1 ÷ 3 = 0.3333...

never ends in decimal.

0.1

never ends in binary.

Computers store numbers in binary — so 0.1 is always slightly off.

XPath: `0.1 + 0.2`

0.30000000000000004

instead of 0.3

XPath: `4.35 * 100`

434.99999999999994

instead of 435

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EXACT INVOICE ARITHMETIC · 2/5

Decimal has been standard since 2008

For 23 years the floating-point standard knew only binary — XSLT still calculates that way by default.

1985	1987	2007	2008	2011
IEEE 754	**IEEE 854**	**IBM POWER6**	**IEEE 754-2008**	**ISO/IEC/IEEE 60559**
Floating-point standard: binary only	any radix, hardly implemented	Decimal arithmetic in hardware	Decimal becomes part of the standard	the same as an international standard

98.6%

of the numbers in commercial databases are decimal or integer

€0.73, not €0.74

5% tax on a €0.70 phone call, calculated in binary. With a million calls: more than €20 wrong.

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EXACT INVOICE ARITHMETIC · 3/5

… and it rounds differently from commerce

Every credit note is affected.

	Standard XSLT	Commercial
Round −2.5	−2	−3
Round 1.005 to two decimals (stored in binary as 1.00499999…)	1	1.01
VAT on a credit note of −€3.50 at 19%	−0.66	−0.67

XSLT: halves always up

Towards +∞ — even for negative amounts.

Commerce & tax law: halves away from zero

Sect. 14.5 UStAE, DIN 1333, AFNOR XP Z12-012

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EXACT INVOICE ARITHMETIC · 4/5

Svanton today, the standard tomorrow

TODAY

Svanton

- Enhanced Saxon, the most widely used free XSLT engine
- Decimal arithmetic, commercial rounding
- Drop-in replacement, tested with the EN 16931 artefacts

→

PROPOSED

XSLT 4.0

- Proposal to the W3C group behind XPath, XQuery and XSLT
- One switch per rule set
- No rule has to change

→

GOAL

Every processor

- Exact invoice arithmetic by default
- Same result everywhere
- Svanton becomes obsolete

UNTIL THEN — FOR EVERYONE WHO WRITES RULES OR CALCULATES INVOICES

calculate in decimal | round commercially | round once — at the end | multiply first, divide last

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EXACT INVOICE ARITHMETIC · 5/5

On the way to the standard

Then nobody needs a special engine any more.

> … I guess we have to give this a shot.
>
> Michael Kay, editor of the XSLT specification and author of Saxon, on the proposal on 28 Sep 2026

What the proposal adds to XSLT 4.0

- Decimal arithmetic for a whole rule set — switched on once
- Commercial rounding (half away from zero) as the default
- No change to existing rules — instead of rebuilding every single rule

Funded by the European Union (StandICT.eu fellowship) · Proposal: github.com/qt4cg/qtspecs/issues/2935

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AMENDMENT · MY PROPOSAL 1/2

Round only once (#163)

Only what is paid gets rounded — nothing else.

Paid amounts — rounded once

- BT-115 Amount due for payment, in the Minor Unit (EUR: cent)
- BT-114 Rounding amount carries the one rounding
- BT-113, BT-172, BT-179, BT-183 likewise — they change the amount due

Everything else, VAT included: as calculated

- Prices, lines, totals, taxable amount and VAT
- Cent VAT only via a national CIUS (e.g. UStAE 14.5 (20))
- Rounding to the Minor Unit today stays valid (warning)

WHAT CHANGES

- 6.5.14 applies to the whole chain, not just per rule
- Exact validation (decimal128): equal → valid, within tolerance → warning, otherwise error
- Today's tolerances remain — they now separate warning from error
- No requirement on software: neither a maximum nor a minimum of decimal places
- Syntax unchanged: amounts in UBL and CII are xsd:decimal

Status 14 Oct 2026: Update 3 in WG1 · warning/error following Cyrille Sautereau's 3rd option · open: DRR with cent lines (FR), backward compatibility

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CALCULATION CHAIN · EN 16931-1:2017

From gross price via net price to VAT

One intermediate rounding: **BT-131 to 2 decimal places**

INPUT & INTERMEDIATE VALUES

BT-146	**Item net price** = BT-148 − BT-147	unlimited unrounded	± 0 stated exactly
BT-131	**Invoice line net amount** = BT-146 × BT-129 ÷ BT-149 + Σ BT-141 − Σ BT-136	2 dp interim rounding	± €0.005 per line
BT-106	**Sum of invoice line net amount** = Σ BT-131	2 dp Σ of rounded values	± €0.50 × 100 lines

FINAL RESULTS

BT-116	**VAT category taxable amount** = Σ BT-131 + Σ BT-99 − Σ BT-92	2 dp interim rounding	± €0.50 Σ of rounded lines
BT-117	**VAT category tax amount** = BT-116 × BT-119 ÷ 100	2 dp interim rounding	± €0.10 €0.50 × 19%
BT-110	**Invoice total VAT amount** = Σ BT-117	2 dp Σ of rounded values	± €0.10
BT-115	**Amount due for payment** = BT-110 + BT-109 Invoice total amount without VAT − BT-113 Paid amount + BT-114 Rounding amount	2 dp final value · bank	± €0.60 €0.50 + €0.10

ALLOWED DECIMAL PLACES

BT-148 Item gross price	unlimited
BT-147 Item price discount	unlimited
BT-146 Item net price	unlimited
BT-129 Invoiced quantity	unlimited
BT-149 Item price base quantity	unlimited
BT-141 Invoice line charge or tax amount	2 dp
BT-136 Invoice line allowance amount	2 dp
BT-131 Invoice line net amount	2 dp
BT-106 Sum of invoice line net amount	2 dp
BT-99 Document level charge or tax amount	2 dp
BT-92 Document level allowance amount	2 dp
BT-119 VAT category rate (law)	unlimited

DECIMAL PLACES · FINAL VALUES

BT-116 VAT category taxable amount	2 dp
BT-117 VAT category tax amount	2 dp
BT-110 Invoice total VAT amount	2 dp
BT-115 Amount due for payment	2 dp

Worst case with number of lines and quantity:
→ "How large does the deviation get?"

Red = rounded intermediate value Green = paid amount, rounded once (bank) Orange = rounded input value
dp = decimal places · 2017 (Tab. 26): all amounts fixed 2 dp, regardless of the currency · ± for 100 lines of 100 units, 19% VAT
Rainbow = order of the calculation steps; a BT keeps its colour where it is reused further down

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CALCULATION CHAIN · EN 16931-1:2026

From gross price via net price to VAT

Two intermediate roundings: **BT-146 to Minor Unit + 2** **BT-131 to the Minor Unit**

INPUT & INTERMEDIATE VALUES

BT-146	**Item net price** = BT-148 − BT-147	Minor Unit + 2 interim rounding	± €0.00005 price rounded once
BT-131	**Invoice line net amount** = **BT-146** × BT-129 ÷ BT-149 + Σ BT-141 − Σ BT-136	Minor Unit interim rounding	± €0.01 €0.005 + 100 × €0.00005
BT-106	**Sum of invoice line net amount** = Σ **BT-131**	Minor Unit Σ of rounded values	± €1.00 × 100 lines

FINAL RESULTS

BT-116	**VAT category taxable amount** = Σ **BT-131** + Σ BT-99 − Σ BT-92	Minor Unit interim rounding	± €1.00 Σ of rounded lines
BT-117	**VAT category tax amount** = **BT-116** × BT-119 ÷ 100	Minor Unit interim rounding	± €0.19 €1.00 × 19%
BT-110	**Invoice total VAT amount** = Σ **BT-117**	Minor Unit Σ of rounded values	± €0.19
BT-115	**Amount due for payment** = **BT-110** + **BT-109** Invoice total amount without VAT − **BT-113** Paid amount + **BT-114** Rounding amount	Minor Unit final value · bank	± €1.19 €1.00 + €0.19

ALLOWED DECIMAL PLACES

BT-148 Item gross price	Minor Unit + 2
BT-147 Item price discount	Minor Unit + 2
BT-146 Item net price	Minor Unit + 2
BT-129 Invoiced quantity	4 dp
BT-149 Item price base quantity	2 dp
BT-141 Invoice line charge or tax amount	Minor Unit
BT-136 Invoice line allowance amount	Minor Unit
BT-131 Invoice line net amount	Minor Unit
BT-106 Sum of invoice line net amount	Minor Unit
BT-99 Document level charge or tax amount	Minor Unit
BT-92 Document level allowance amount	Minor Unit
BT-119 VAT category rate (law)	2 dp

DECIMAL PLACES · FINAL VALUES

BT-116 VAT category taxable amount	Minor Unit
BT-117 VAT category tax amount	Minor Unit
BT-110 Invoice total VAT amount	Minor Unit
BT-115 Amount due for payment	Minor Unit

Worst case with number of lines and quantity:
→ "How large does the deviation get?"

Red = rounded intermediate value **Green** = paid amount, rounded once (bank) **Orange** = rounded input value
dp = decimal places · Minor Unit = smallest currency unit, ISO 4217 (EUR: 2 dp) · ± for 100 lines of 100 units, 19% VAT
Rainbow = order of the calculation steps; a BT keeps its colour where it is reused further down

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CALCULATION CHAIN · PROPOSED EN 16931-1:2026 AMENDMENT (#163)

From gross price via net price to VAT

No intermediate rounding — only BT-115 Amount due for payment is rounded; BT-114 Rounding amount shows it.

INPUT & INTERMEDIATE VALUES

BT-146	**Item net price** = BT-148 − BT-147	free unrounded	± 0 as agreed
BT-131	**Invoice line net amount** = BT-146 × BT-129 ÷ BT-149 + Σ BT-141 − Σ BT-136	free unrounded	± 0 exact
BT-106	**Sum of invoice line net amount** = Σ BT-131	free unrounded	± 0 exact sum

FINAL RESULTS

BT-116	**VAT category taxable amount** = Σ BT-131 + Σ BT-99 − Σ BT-92	free unrounded	± 0 exact
BT-117	**VAT category tax amount** = BT-116 × BT-119 ÷ 100	free unrounded	± 0 Minor Unit only via CIUS → warning
BT-110	**Invoice total VAT amount** = Σ BT-117	free unrounded	± 0
BT-115	**Amount due for payment** = BT-110 + BT-109 Invoice total amount without VAT − BT-113 Paid amount + BT-114 Rounding amount	Minor Unit rounded once	± 0 BT-114 shows the rounding

ALLOWED DECIMAL PLACES

BT-148 Item gross price	free
BT-147 Item price discount	free
BT-146 Item net price	free
BT-129 Invoiced quantity	free
BT-149 Item price base quantity	free
BT-141 Invoice line charge or tax amount	free
BT-136 Invoice line allowance amount	free
BT-131 Invoice line net amount	free
BT-106 Sum of invoice line net amount	free
BT-99 Document level charge or tax amount	free
BT-92 Document level allowance amount	free
BT-119 VAT category rate (law)	2 dp

DECIMAL PLACES · FINAL VALUES

BT-116 VAT category taxable amount	free
BT-117 VAT category tax amount	free
BT-110 Invoice total VAT amount	free
BT-115 Amount due for payment	Minor Unit

Worst case with number of lines and quantity:
→ "How large does the deviation get?"

Red = rounded intermediate value Green = paid amount, rounded once (bank) Orange = rounded input value
dp = decimal places · Minor Unit = smallest currency unit, ISO 4217 (EUR: 2 dp) · ± for 100 lines of 100 units, 19% VAT
Rainbow = order of the calculation steps; a BT keeps its colour where it is reused further down

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WORST CASE · EN 16931-1:2017

How large does the deviation get?

One factor: the number of lines n — each line rounds **BT-131** Invoice line net amount and loses up to ± €0.005.

Factor 1 · price rounding × quantity Q

BT-146 Item net price is unlimited in 2017 — the price is not rounded, so the quantity magnifies no error.

`none`

Factor 2 · number of lines n

Each line rounds **BT-131** Invoice line net amount to 2 decimal places: up to ± €0.005. **BT-106** Sum of invoice line net amount and **BT-116** VAT category taxable amount add up all n lines — and with them the errors of all lines.

`± n × €0.005`

NET DEVIATION **BT-106** Sum of invoice line net amount / **BT-116** VAT category taxable amount

Units per line (Q) → ↓ Lines (n)	100 units	10,000 units	1,000,000 units
1 line	Price × quantity ± 0 + rounding BT-131 ± €0.005 = ± €0.005	Price × quantity ± 0 + rounding BT-131 ± €0.005 = ± €0.005	Price × quantity ± 0 + rounding BT-131 ± €0.005 = ± €0.005
100 lines	± €0.50	± €0.50	± €0.50
10,000 lines	± €50.00	± €50.00	± €50.00

Assumption: all roundings go in the same direction, BT-149 Item price base quantity = 1, 19% VAT.

TOTAL · NET

n × €0.005

VAT: 19% of that. Open-ended — the more lines, the larger the deviation.

100 lines of 10,000 units
Net ± €0.50 VAT ± €0.10

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WORST CASE · EN 16931-1:2026

How large does the deviation get?

Two factors: first the quantity Q × rounded **BT-146** Item net price, then the number of lines n — open-ended.

Factor 1 · price rounding × quantity Q

BT-146 Item net price is "squeezed" once into Minor Unit + 2 (EUR: 4 dp): up to ± €0.00005. **BT-131** Invoice line net amount multiplies this one error by **BT-129** Invoiced quantity. #148: €0.123449 → €0.1234; with 10,000 units €0.49 is missing.

`± Q × €0.00005 per line`

Factor 2 · number of lines n

Each line rounds **BT-131** Invoice line net amount additionally to the Minor Unit: up to ± €0.005. **BT-106** Sum of invoice line net amount and **BT-116** VAT category taxable amount add up all n lines — and with them the errors of all lines.

`± n × €0.005`

NET DEVIATION **BT-106** Sum of invoice line net amount / **BT-116** VAT category taxable amount

Units per line (Q) → ↓ Lines (n)	100 units	10,000 units	1,000,000 units
1 line	Price × quantity ± €0.005 + rounding BT-131 ± €0.005 = ± €0.01	Price × quantity ± €0.50 + rounding BT-131 ± €0.005 = ± €0.505	Price × quantity ± €50.00 + rounding BT-131 ± €0.005 = ± €50.005
100 lines	± €1.00	± €50.50	± €5,000.50
10,000 lines	± €100.00	± €5,050.00	± €500,050.00

Assumption: all roundings go in the same direction, BT-149 Item price base quantity = 1, 19% VAT. Likewise: BT-129 Invoiced quantity only 4 dp — up to ± 0.00005 × price per line for expensive items.

TOTAL · NET

n × (Q × €0.00005 + €0.005)

VAT: 19% of that. Open-ended — more lines or larger quantities, more deviation.

100 lines of 10,000 units
Net **± €50.50** VAT **± €9.60**

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WORST CASE · PROPOSED AMENDMENT (#163)

How large does the deviation get?

No factor: without intermediate rounding it stays at ± 0 — for any quantity and any number of lines.

Factor 1 · price rounding × quantity Q

BT-146 Item net price and **BT-131** Invoice line net amount stay exact — the quantity magnifies no error.

none

Factor 2 · number of lines n

BT-131 Invoice line net amount stays exact — it does not matter how many lines **BT-106** Sum of invoice line net amount and **BT-116** VAT category taxable amount add up.

none

NET DEVIATION **BT-106** Sum of invoice line net amount / **BT-116** VAT category taxable amount

Units per line (Q) → ↓ Lines (n)	100 units	10,000 units	1,000,000 units
1 line	Price × quantity ± 0 + rounding BT-131 ± 0 = ± 0	Price × quantity ± 0 + rounding BT-131 ± 0 = ± 0	Price × quantity ± 0 + rounding BT-131 ± 0 = ± 0
100 lines	± 0	± 0	± 0
10,000 lines	± 0	± 0	± 0

Assumption: all roundings go in the same direction, BT-149 Item price base quantity = 1, 19% VAT.

TOTAL · NET

± 0

Only BT-115 Amount due for payment is rounded once; BT-114 Rounding amount shows it.

100 lines of 10,000 units
Net ± 0 VAT ± 0

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AMENDMENT · MY PROPOSAL 2/2

Gross prices for the EN core (#164)

If the price is fixed including VAT, it should lead the invoice.

Step	Net model (today)	Gross model (new)
Price	BT-146 Item net price	BT-146V Item net price plus VAT
Line	BT-131 = price × quantity …	BT-131V = price plus VAT × quantity …
Per category	BT-116 = Σ BT-131 …	BT-116V = Σ BT-131V …
VAT	BT-117 = BT-116 × rate added	BT-117 = BT-116V × rate ÷ (100 + rate) extracted
Net per category	(given)	BT-116 = BT-116V − BT-117

The model is selected via BT-146V on each line — no indicator needed. BT-146 remains mandatory (Art. 226(8) VAT Directive), in the gross model for information only. Not for categories N, D, F, I, J.

Often impossible today: At 19% VAT, 16% of all gross amounts cannot be reached with net and VAT in the Minor Unit — €9.99 = 8.39 + 1.59 → 9.98. With 5 lines 52%, with 35 lines 81% miss the receipt total.

EXAMPLE: FIXED BOOK PRICES

1,000 books at €24.95 incl. 7% VAT

	Total	VAT
Target	24,950.00	1,632.24
Net 2026	24,950.05	1,632.25
Gross model	24,950.00	1,632.24

Net model: 5 cents too much, VAT 1 cent too high. Gross model: VAT €1,632.24299… → €1,632.24 only due to the cent rule (UStAE) — #163: warning.

Use cases: **fixed book prices · receipt → invoice · retail and marketplaces · statutory gross prices**

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STATUS QUO

Current status

Revision of the standard

- March 2026: final vote on EN 16931-1 completed
- May 2026: final vote on the syntax bindings (TS 16931-3) completed
- German edition DIN EN 16931-1:2026-10 now available

New syntaxes: UBL 2.5 & CII D25A

- UBL 2.5: OASIS Standard since 12 August 2026
- CII D25A: published by UN/CEFACT in June 2026
- Next step: the European Commission adds both to the list of syntaxes

Validation

- Work interrupted by open copyright questions (IP problem)
- Derivative use hopefully possible soon
- EU validator on GitHub and e-Invoice Advisory Group have expired — progress unclear



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STATUS · STANDARD

EN 16931-1:2026

Adopted at European level — the German edition has been published.

Adopted & available

- Ratified by CEN on 13 Mar 2026
- English edition since 18 Mar 2026 (DAV)
- DIN EN 16931-1:2026-10 available from DIN Media

Why the DIN edition came later

- CCMC (CEN-CENELEC Management Centre) dropped the 16-week translation phase — under pressure from the European Commission, in favour of the English edition
- In addition, more than 40 pages of comment tables to discuss and incorporate

CEN MILESTONES

- **2026-03-13** Ratification
- **2026-03-18** Available · English edition (DAV)
- **2026-06-30** National announcement (DOA)
- **2026-09-30** **German edition published (DOP)**
- **2029-03-31** Withdrawal of EN 16931-1:2017 (DOW)

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SYNTAX BINDING · NEXT STEPS

Work at CEN on an amendment / corrigendum

ISSUE TRACKER

64

Issues for the CEN members on GitHub since 2020

Contributions from DE, FR, NL and the OASIS UBL community

Current findings in the syntax bindings (September 2026)

- CII: the VAT exchange rate is bound to the wrong element
- CII: a line-level delivery location is bound to the invoice header
- Checks for “true/false” miss the equally valid values “1/0”
- …

Proposal: a corrigendum for the syntax bindings of UBL and CII — so that every path is unambiguous and machine-readable

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OUTLOOK · SEMANTIC AMENDMENT

Where is the journey going?

Revision and syntax binding are published — work on the amendment of the semantics has started.

AMENDMENT

~50

Topics under discussion

- Code list revision in progress
- Goal: completion by end of 2026
- Syntax binding in parallel where possible

Tax & DRR

- Exemption reason for categories besides E?
- Goods/services, subtotals and DRR
- Third-party fees, amount due and DRR
- VAT group as one taxable person

Business processes

- Multiple payment terms with date and amount
- Incoterms
- Credit note: standalone or corrective, negative credit notes
- Replacing corrective invoice

Technology & interoperability

- BT-10 Buyer reference and backward compatibility
- Split BT-161 into value and measurement
- Exchange rate: notation and operator
- Spaces, CR/LF, tab in identifiers
- Different CIUS at buyer and seller
- Extensions in cross-border ViDA traffic

My contributions to the amendment: **#163 Round only once · #164 Gross price model**

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Time for your questions

Discussion · exchange · thinking through next steps together

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CONTACT

Talk to me

Slides as PPTX

Svante Schubert
Syntax bindings · validation · tooling
Editor of the syntax bindings, CEN/TC 434 WG3

Follow & join in

- WG1 issues #163 and #164
- Issues on the syntax bindings (WG3)
- Svanton: exact arithmetic for Saxon
- XSLT 4.0 proposal (qt4cg #2935)

Thank you for your attention — I look forward to the exchange at the ZUGFeRD-Entwicklertage.

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